

Arrival choices

Previous lesson reminder:

Start with 1 and 2. Try 3 and 4 when ready. Reading and exact scribing are available.

1 Which units belong on the temperature axis?

Support: Use the reminder strip or talk it through with an adult.

My response

2 Which meaning fits tectonic?

Support: Choose: Related to movement of Earth's plates OR People or things in harm's way.

My response

3 What makes the report more than a description of the event?

Support: Use the model evidence anchor assessment or read one relevant card together.

My response

4 What must accompany the graph or map?

Support: Give a reason when ready.

My response

Stretch: use a precise example or explain which detail supports your answer.

Evidence cards

Read, point or listen. Keep the source label with any detail you use.

A Japan 11 March 2011

A magnitude 9.1 earthquake occurred off north-east Honshu, Japan. Sudden movement on the subduction boundary displaced the seafloor and produced a tsunami.

Source: USGS event record, official20110311054624120_30

Limit: Magnitude measures the earthquake; it is not a direct count of people affected.

B Effects and responses

Shaking and coastal flooding damaged places and disrupted people's lives. Earthquake-resistant design addresses shaking; warnings, evacuation routes and higher ground address tsunami exposure.

Source: Authored explanation of the case and risk-management principles

Limit: A sea wall or warning cannot eliminate all risk. Do not judge people without considering access and warning time.

C Heat and unequal risk

On 19 July 2022 the Met Office recorded 40.3°C at Coningsby, 40.2°C at St James's Park and 39.6°C at Topcliffe. Outdoor work increases exposure; difficulty accessing cool spaces can increase vulnerability. Shade and accessible cool spaces can reduce harm.

Source: Met Office July 2022 report, station table; risk explanation authored for teaching

Limit: This explains risk factors, not a medical assessment of an individual.

D Reading a hazard feed

A USGS event row can include UTC time, place, magnitude and depth in kilometres. A short feed is a dated snapshot. It cannot predict the next earthquake.

Source: USGS earthquake feed documentation

Limit: Small samples, reporting thresholds and later revisions affect patterns.

Visual resource

Use this visual with the evidence cards and enquiry questions.



Source: Natural Earth public-domain outlines, labels simplified. Terms of use:
<https://www.naturalearthdata.com/about/terms-of-use/> (recorded in Sources and checks).

A detail I noticed and what it helps me explain

Data for the enquiry

Selected station maxima on 19 July 2022

Station	Maximum °C
Coningsby	40.3
St James's Park	40.2
Topcliffe	39.6
Sheffield	39.4

Units: °C

Source: Met Office July 2022 heatwave report, table on page 8

Limit: Four selected station maxima are not a national average or a long-term climate trend. The report marked values provisional at publication.

My labelled graph or data notes

Shared investigation

1. Audit a practice report against location, causes, effects, risk and management.

2. Choose a useful map or graph and explain why it belongs.

Work with a partner or adult, or use your own table. One person suggests; the other checks the evidence. Swap after one turn. Passing is allowed.

Use the evidence cards and any visual sheet. Follow the two shared steps above, then record one decision.

Our explanation

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining.

Further challenge: Weigh two management approaches and explain how the evidence supports your final judgement.

Supported independent task

Complete this route only. Complete a hazard case-study report.

1. Produce an independent case-study report for Japan 2011 or the July 2022 heatwave.
2. Include location, causes, effects, risk, management and a useful map or graph.
3. Justify a judgement and evaluate a source or data limitation.

Support: Use the glossary and one chunk of evidence. Plan the claim and supporting detail before explaining. Complete the organiser one space at a time. Location _____. Cause _____. Effects _____. Exposure and vulnerability _____. Management judgement _____. Evidence limit _____.

My answer to the assessment enquiry

My evidence and source

My explanation or comparison

My conclusion and what the evidence does not show

An adult may read or scribe your exact response. They should not choose the answer.

Standard independent task

Complete this route only. Complete a hazard case-study report.

1. Produce an independent case-study report for Japan 2011 or the July 2022 heatwave.
2. Include location, causes, effects, risk, management and a useful map or graph.
3. Justify a judgement and evaluate a source or data limitation.

Support: Location _____. Cause _____. Effects _____. Exposure and vulnerability _____. Management judgement _____. Evidence limit _____.

My answer to the assessment enquiry

My evidence and source

My explanation or comparison

My conclusion and what the evidence does not show

Check the required evidence and revise one detail.

Stretch independent task

Complete this route only. Complete a hazard case-study report.

1. Produce an independent case-study report for Japan 2011 or the July 2022 heatwave.
2. Include location, causes, effects, risk, management and a useful map or graph.
3. Justify a judgement and evaluate a source or data limitation.
4. Weigh two management approaches and explain how the evidence supports your final judgement.

Support: Choose the evidence independently. Use the frame only if it helps.

My answer to the assessment enquiry

My evidence and source

My explanation or comparison

My conclusion and what the evidence does not show

Weigh two management approaches and explain how the evidence supports your final judgement.

Exit ticket

Choose a response mode. You may give your answers privately to an adult.

What makes the report more than a description of the event?

My response

What must accompany the graph or map?

My response

Support: Location _____. Cause _____. Effects _____. Exposure and vulnerability _____. Management judgement _____. Evidence limit _____.

Further challenge: Weigh two management approaches and explain how the evidence supports your final judgement.

One thing I changed or want to check
